

Igor Farias Campos

Offensive Security Researcher (Apple Platforms)

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Profile

Offensive Security Researcher specializing in the Apple ecosystem (iOS/macOS), with a core focus on Kernel vulnerability research and OS Internals. Dedicated to advanced reverse engineering and deep system analysis, I have a proven track record of identifying critical vulnerabilities, highlighted by the discovery of **CVE-2025-43534** (iOS Sandbox Escape). Beyond exploit research, I develop custom low-level solutions for advanced platform security analysis.

Awards

03/22/2025	Bsides Rio de Janeiro CTF — 1st Place • Solved advanced challenges in binary exploitation, reverse engineering, and vulnerability analysis. • Performed static and dynamic analysis of stripped binaries, identifying memory corruption primitives and logic flaws. • Developed custom exploits leveraging stack/heap corruption, function pointer overwrites, and control-flow manipulation. • Used tools such as Ghidra, LLDB, and custom scripts to reverse engineer challenge binaries and bypass mitigations. • Demonstrated strong problem-solving under time constraints, collaborating effectively in an offensive security setting.
10/21/2025	CVE-2025-43534  <i>iOS Sandbox Escape (itunesstored Arbitrary File Write)</i> • Discovered and weaponized a logical vulnerability in iOS download daemons, achieving an Arbitrary File Write outside the application sandbox. • Engineered a 2-stage exploit chain by crafting malicious SQLite databases (downloads.28.sqlitedb), manipulating the itunesstored daemon to bypass path validation constraints. • Successfully chained the vulnerability with EPUB extraction processes to write arbitrary files across the iOS Data partition (/private/var/). This granted unauthorized modification of restricted mobile-owned directories, system preferences, and caches (e.g., FairPlay, SystemGroup), enabling high-impact compromises such as physical Activation Lock bypass.
04/20/2025	Open-Source Security Contribution — iometa  <i>tool by Siguza</i> • Identified and debugged a kernelcache parsing crash in <i>iometa</i> when analyzing iOS 15 and earlier kernelcaches. • Performed LLDB-based root-cause analysis of an EXC_BAD_ACCESS caused by invalid Mach-O fixup chain assumptions across kernelcache layout versions. • Fixed a null pointer dereference in macho_validate_fixup_chain by correcting version-specific structure handling. • Patch was merged upstream, restoring compatibility with older kernelcaches; contribution acknowledged by the project author.

Professional Experience

05/2025 – Present Brazil	Mobile Security Engineer <i>iFood</i>  • iFood is Brazil's leading delivery platform, serving ~55M active users, operating across 1,500+ cities, and processing ~120M orders per month (with a reported record of 180M orders in November 2025). • Hired into a dedicated Mobile Security organization to strengthen iOS security posture at scale for a high-traffic consumer platform. • Worked on iOS internals-driven security engineering, combining reverse engineering and static analysis to validate hardening assumptions and detect risky build / packaging deviations before release. • Built security automation for CI/CD so mobile security requirements become enforceable controls (deterministic pass/fail behavior, consistent diagnostics, auditability) across multiple app pipelines. • Focused on preventing common high-impact issues such as shipping sensitive assets, missing protection components, or misconfigured security-critical settings in production builds.
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02/2024 – 05/2025
Brazil

iOS Developer

Apple Developer Academy 

- At the **Apple Developer Academy**, I developed over 5 apps per year, all written in Swift, utilizing a **wide range of frameworks**.
- Maintained my previous apps by utilizing tools such as **PostHog** for analytics and **App Store Connect** for app management, achieving **over 10,000 impressions** and **180+ downloads**, while ensuring smooth performance and user satisfaction.
- Participated in a comprehensive design project, leading the process from prototyping to full implementation, gaining practical experience in crafting user-centered solutions.

11/2024 – 05/2025
Brazil

iOS Security Researcher

MAD 

- Serve as the lead researcher on the functionality of intrusion tools, ensuring the team stays ahead of emerging threats and vulnerabilities.
- Developed and contributed **over 10 detection** functions for identifying issues such as jailbreaks, sandbox escapes, and tools like Frida, enhancing the robustness of our **RASP solution**.

2020 – 2023
Brazil

Independent Hardware Repair Technician

Self-employed

- Performed board-level diagnostics and micro-soldering on mobile devices, analyzing power, signal, and component-level failures.
- Developed strong intuition for hardware–software interaction, fault isolation, and systematic debugging under real-world constraints.

Projects

MAD (Mobile Application Defense)

iOS RASP & Threat Intelligence Engine

- **Protected enterprise banking applications at scale** by architecting the core iOS Runtime Application Self-Protection (RASP) framework, monitoring thousands of daily security events and mitigating 50+ platform-specific vulnerabilities.
- **Defeated modern instrumentation and rootless jailbreaks** (Frida, Dopamine, RootHide) by engineering real-time detection mechanisms via pre-main `attribute((constructor))` initialization, raw XNU syscalls (`SYS_csops`), and Objective-C runtime introspection.
- **Enforced low-level memory and execution integrity** by implementing ARM64 trampoline pattern detection, Mach-O load command parsing, `vm_protect` manipulation, and querying undocumented XPC bootstrap pipes to thwart sandbox escapes, dylib injections, and API hooking.

SBPL Viewer

Apple Sandbox Internals Analysis Tool

- **Engineered a custom parser for Apple's undocumented Sandbox Profile Language (SBPL)**, enabling deep static analysis of iOS/macOS sandbox boundaries, IPC restrictions, and allowed system calls.
- **Accelerated Sandbox Escape research** by translating compiled, proprietary binary profiles into human-readable formats, allowing rapid identification of misconfigured path validations and privileged daemon interactions.

KnightWatch

Rust-based Mobile Binary Analyzer & CI/CD Security Gate

- **Automated enterprise security validation at scale** by architecting a Dockerized, Rust-based SAST engine for GitLab CI/CD pipelines, strictly gating IPA/APK releases and cross-validating RASP obfuscation claims (DexProtector, Promon) prior to production deployment.
- **Implemented deep low-level Mach-O inspection**, parsing ARM64 PAC instructions, dyld4 chained fixups, and Objective-C runtime structures (`class_ro_t`, `__objc_classlist`) to detect encryption failures, leaked class names, and bypassed runtime protections.
- **Enforced zero-trust deployment policies** by calculating Shannon entropy for encryption verification, auditing binary hardening (PIE, ARC, NX), and analyzing entitlements and CMS/PKCS#7 provisioning profiles to generate quantified security scores and deterministically block vulnerable builds.

iBootWatchOS32Patcher

Custom Boot-Chain Exploit Tooling

- **Unlocked runtime security research** for all ARM32 Apple Watch devices (S1–S3) by engineering a universal binary patcher that defeats the Apple IMG4 Secure Boot chain (watchOS 2.x–8.x) at the iBSS/iBEC bootloader stages.
- **Automated cryptographic bypasses** by implementing ARM32 Thumb-2 instruction-level pattern scanning, dynamically patching IMG4 signature validation callbacks in-place, and parsing Mach-O/FAT headers for kernelcache TrustCache manipulation.
- **Accelerated vulnerability research workflows**—reducing boot-chain reverse engineering effort from days to minutes—by automating boot-args injection, DeviceTree tag manipulation, and code-signing (AMFI) enforcement disablement.

Languages

English — Conversational

Portuguese — Native/Bilingual

Spanish — Basic